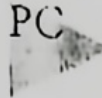


Answer all Questions

Provide appropriate illustrations and assume relevant values wherever necessary.

Q1	Write a piece of code that consists of at least ten instructions, where four of them are independent, and the remaining are dependent. For this code: - Propose the number of scalar processors you would use. - Prove that the number of proposed processors is optimal and takes the least execution time. - Suggest a different type of processor than the scalar that could further reduce execution time, and justify your answer. - Calculate the maximum number of processors and computing resources utilized for c) Assumption: Assign a maximum of two threads per processor.	10
Q2	Consider A and B, each containing 256 data elements. These are used in multiple operations: $C = A + B$ $D = A \times B$ $E = A - B$ $F = (A + B) \times (A - B)$ The processor's register can hold up to 64 elements at a time. - Propose a suitable processor type for this workload, and justify your choice. - Design the processor for the above operations. Support your design with a suitable block diagram. - Consider that the above 4 operations have increased to 4,000 and perform on a billion data elements. For this scenario, design a parallel system internal-thread level block diagram and briefly explain how these operations are performed at the thread level using only a diagram.	15
Q3	Each thread executes 3 arithmetic instructions, each with a 4-cycle latency, followed by a memory load. Two types of load occur: a cache hit, which takes 6 cycles, and a cache miss, which takes 14 cycles. These two types occur equally often across the thread population. - Calculate the total loop time for one thread, for both the cache-hit and cache-miss cases. Using these, calculate the minimum number of threads for 100% execution-unit utilization, and show the core utilization at your calculated thread count and at one smaller thread count. - Propose a suitable processor architecture that can hide these per-thread waiting periods.	8

Q4	For the following instruction streams, propose and draw the suitable processor architecture. Instruction stream 0 <pre> vector_ld v0, vector_addr[r1] vector_mul v1, v0, v0 add r2, r1, r0 vector_add v2, v0, v0 vector_mul v3, v1, v2 vector_st addr[r2], v0 </pre>  Instruction stream 1 <pre> ld r0, addr[r1] mul r1, r0, r0 add r2, r1, r0 mul r3, r1, r2 st addr[r2], r0 </pre>  Instruction stream 2 <pre> ld r0, addr[r1] mul r1, r0, r0 add r2, r1, r0 mul r3, r1, r2 st addr[r2], r0 </pre> 	7
Q5	Assume a processor node frequently processes both a local and a neighbouring weather grid without requesting permission to access data from its neighbours. - Propose a memory architecture that minimizes latency and maximizes throughput. - Draw the weather grid and connect them with suitable interconnection network that would reduce communication bottlenecks while keeping costs reasonable? Justify your answer. - If a local node needs permission to access data from its neighbours, recommend an appropriate memory architecture.	10